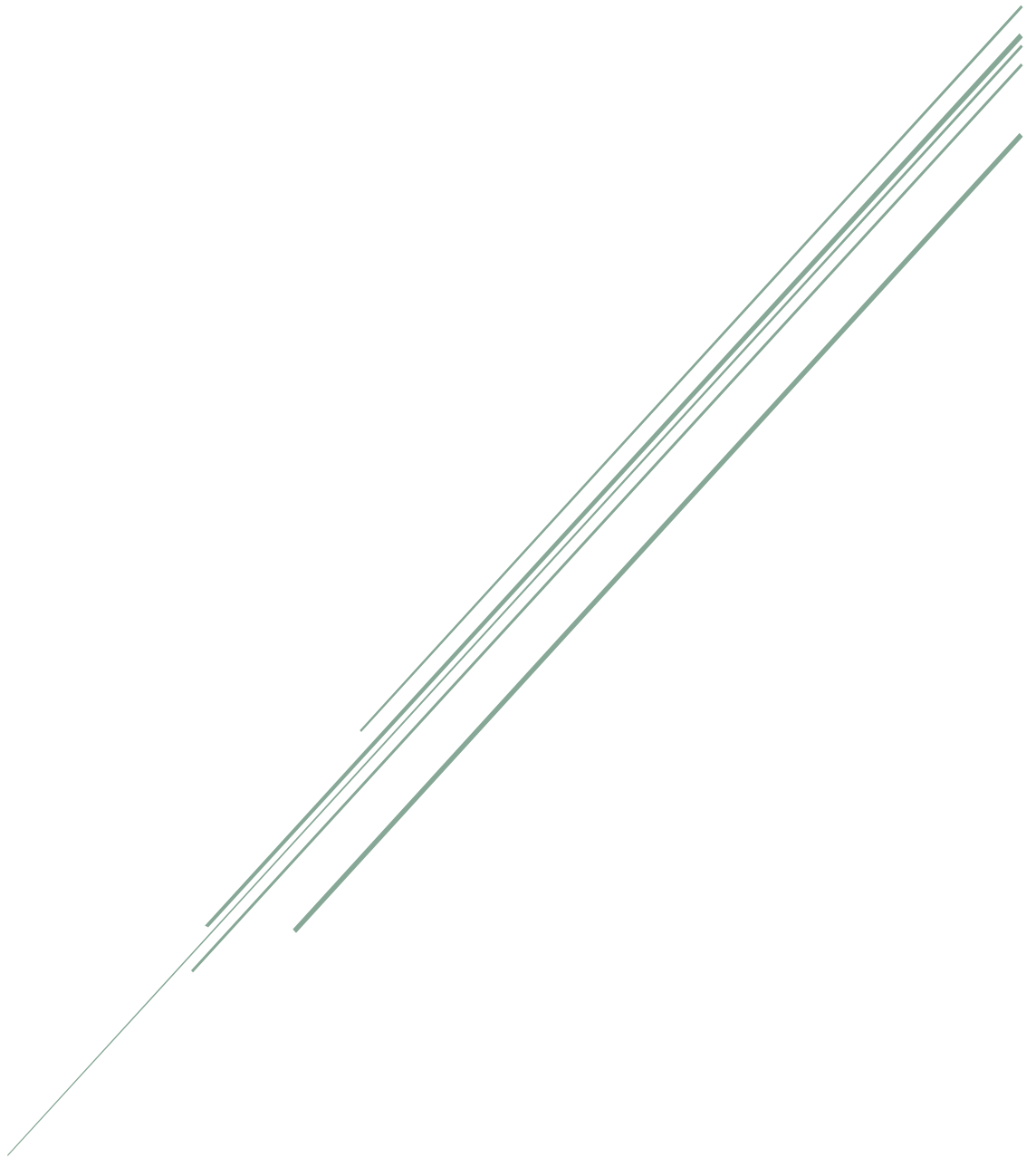


WORLD SERPENT INN

Technical Documentation



ISPW project made by:
Ivan Crescenzi (0322029) and Arianna Gabrieli (0323240)

INDEX

1. SOFTWARE REQUIREMENTS SPECIFICATION

1.1. INTRODUCTION

- 1.1.1. Aim of the document
- 1.1.2. Overview of the defined system
- 1.1.3. Hardware and software requirements
- 1.1.4. Related systems, PROs and CONs

1.2. USER STORIES

(Arianna Gabrieli)

- 1.2.1. Campaign participation request (Player)
- 1.2.2. Campaign selection (Player)
- 1.2.3. Search with filter (Player)

(Ivan Crescenzi)

- 1.2.4. Campaign frequency (DM)
- 1.2.5. Invite Players (DM)
- 1.2.6. Participation request notification (Dungeon Master)

1.3. FUNCTIONAL REQUIREMENTS

(Arianna Gabrieli)

- 1.3.1. Show available campaigns
- 1.3.2. Notification of new available campaign
- 1.3.3. List of participation requests

(Ivan Crescenzi)

- 1.3.4. Campaign location
- 1.3.5. Prevent creation
- 1.3.6. Username already taken

1.4. USE CASES

- 1.4.1. Overview diagram
- 1.4.2. Internal Steps

2. STORYBOARDS

- 2.1 Login (without info)
- 2.2 Login (with info)
- 2.3 Homepage (user not logged-in)
- 2.4 Homepage (user logged-in)
- 2.5 Character list
- 2.6 Campaign list
- 2.7 Single Campaign view

3. DESIGN

3.1. **CLASS DIAGRAM USE CASES**

- 3.1.1. View of Participating Class (VOPC)
- 3.1.2. Design-level diagram

3.2. **DESIGN PATTERNS**

- 3.2.1. Abstract factory pattern (and Singleton pattern).
- 3.2.2. Observer pattern (and Singleton pattern).

3.3. **ACTIVITY DIAGRAM**

- 3.3.1 Activity Diagram (Use case “Plan campaign participation”).
- 3.3.2 Activity Diagram (Use case “Create campaign”).

3.4. **SEQUENCE DIAGRAM**

- 3.4.1 Sequence Diagram (Use case “Plan campaign participation”).
- 3.4.2 Sequence Diagram (Use case “Create campaign”).

3.5. **STATE DIAGRAM**

- 3.5.1 State Diagram (Use case “Plan campaign participation”).
- 3.5.2 State Diagram (Use case “Create campaign”).

4. TESTING

- 4.1 LoginControl Test (Arianna Gabrieli).

- 4.2 Username Taken Test (Arianna Gabrieli).
- 4.3 CampaignParticipationControl Test (Arianna Gabrieli).
- 4.4 NotificationControl Test (Ivan Crescenzi).
- 4.5 RequestAlreadySent Test (Ivan Crescenzi).

5. DATABASE

6. CODE & SONARCLOUD

7. VIDEO

1.1 SOFTWARE REQUIREMENTS SPECIFICATION

Introduction

1.1.1 Aim of the document

The aim of this document is to precisely describe the “World Serpent Inn” system. Therefore, all internal and external system aspects will be analysed using suitable diagrams.

“World Serpent Inn” is an academic project developed for the “Software Engineering and Web Development” course during the academic year 2025/2026.

The project was made by students Arianna Gabrieli and Ivan Crescenzi.

1.1.2 Overview of the defined system

The designed system is a platform that allows users to plan and manage their participation in “Dungeon & Dragons” game campaigns.

There are three types of users:

- Guest user:
 - *Guest users do not need to be logged-in: they can view and download the character sheets* of a list of public characters, posted by users logged in as Players, to take inspiration from.*
- Player:
 - Players can search through a list of available campaign. To simplify the search, they can filter campaigns by name and mode.
 - Once selected a campaign, Players can submit a participation request.
 - Players officially join a campaign when the respective DM accepts their request.
 - *To help DMs craft unique experiences, Players can create their own characters and upload the respective character sheet on a dedicated section of the website.*
- DM:
 - DMs can create campaigns, accept or reject participation requests from Players and invite specific Players to participate in their campaign at the moment of campaign creation.

Once a Player joins a campaign, an e-mail with participation info (a link to an online platform if the campaign's mode is online or a link to a physical address if the campaign's mode is offline) will be sent to them.

Dictionary:

***Character sheet:** a file containing all relevant character information (name, level, class, race, equipment).

Important notice:

Functionalities written in *italics* above have not been implemented.

1.1.3 Hardware and Software Requirements

Systems Requirements:

- **Operating system:** Windows 10/11, macOS 12+, Linux (Ubuntu 22.04+);
- **CPU:** Dual Core 2.0 GHz or higher;
- **RAM:** minimum 4 GB, recommended 8 GB;
- **Disk Space:** 500 MB – 1 GB of free space;
- **Network:** Broadband Internet Connection required for database hosting.

Software for development:

- **IDE:** IntelliJ IDEA Community Edition;
- **UML Design:** draw.io;
- **Graphic Interface:** Java FX;
- **Database:** MySQL (via Workbench);
- **Language & Build:** Java SDK 25.0.3;

1.1.4 Related systems [PROs & CONs]

D&D Beyond:

- **PROs:** it is extremely advanced in characters creation and official rule management.
- **CONs:** it lacks intuitive, built-in matchmaking tools: users are forced to browse chaotic text forms, without the possibility of filtering their search results.

StartPlaying:

- **PROs:** search filters work very well and, since players pay to participate, it's very unlikely that they will leave campaigns without finishing them. Dropout rate is, consequently, low.
- **CONS:** players must pay to participate in most campaigns: participation is not accessible to everyone.

1.2 USER STORIES

1.2.1 Campaign participation request [Player]

As a Player, I want to send a request of participation to an available campaign*, **so that** I can be part of the game group.

1.2.2 Campaign selection [Player]

As a Player, I want to search through a list of available campaigns, **so that** I can choose which I like the most.

1.2.3 Search with filter [Player]

As a Player, I want to filter campaigns by name and mode, **so that** I can find the campaign I'm looking for faster.

1.2.4 Campaign Frequency [DM]

As a DM, I want to choose how often game sessions will run when I create a campaign*, **so that** a Player who wants to join can know that before joining.

1.2.5 Invite Players [DM]

As a DM, I want to invite players to my campaign as soon as I create it, **so that** the players I've invited can immediately choose whether to participate.

1.2.6 Participation request notification [DM]

As a DM, I want to receive a notification when one of the Players requests to join my campaign, **so that** I can choose whether to accept it or not.

Dictionary

***Campaign** = a series of connected adventures or game sessions played by a consistent group of players and characters within a persistent game world over time.

***Available campaign** = a campaign in which maximum number of players has not been reached yet.

1.3 FUNCTIONAL REQUIREMENTS

1.3.1 Show available campaigns

The system shall display available campaigns in a list to the Players.

1.3.2 Notification of new available campaign

The system shall notify Players selected by a DM when a new campaign is created and published. The notification, found in a dedicated page, shall contain the following text: “You’ve been selected to participate in (campaign name) by (DM’s name)! Do you want to accept?”.

1.3.3 List of participation requests

The system shall display to DMs the list of participation requests from Players to their campaign in a dedicated section.

1.3.4 Campaign location

The system shall display the city in which the campaign takes place if it is offline and the platform if it is online.

1.3.5 Prevent creation

The system shall prevent Players from starting the procedure which creates a campaign by not allowing them to enter the "create campaign" section.

1.3.6 Username already taken

The system shall prevent the user from choosing a name already chosen by another user when registering.

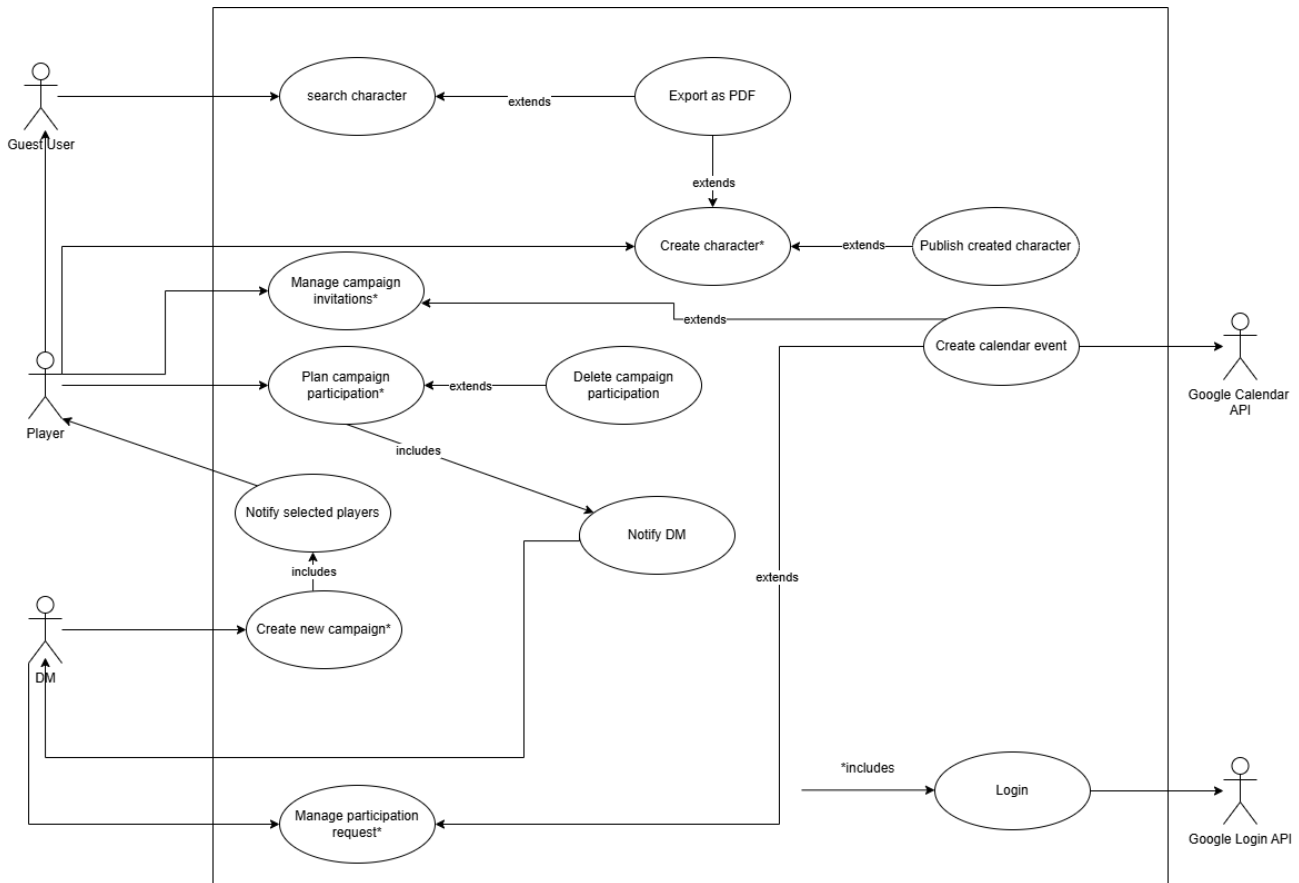
Dictionary

***Online campaign** = a campaign played remotely through communication apps and virtual tabletop platforms.

***Offline campaign** = a campaign where the DM and the Players physically meet in a specific location.

1.4 USE CASES

1.4.1 Overview diagram



1.4.2 INTERNAL STEPS

(Arianna Gabrieli)

“Plan campaign participation”. Precondition: user logged-in as Player.

1. The system retrieves all available campaigns from database.
2. The User applies a search filter*.
3. The system displays the list of filtered campaigns.
4. The User selects a specific campaign.
5. The system displays the details of the selected campaign.
6. The User submits a participation request.
7. The system saves the request and sends it (via system notification) to the DM who created the campaign.

8. The system notifies the User saying that the request had been sent and it is waiting to be approved.
9. The system redirects the User to the homepage.

Extensions

- 1.a: “There are no available campaigns”: the system displays a message informing the user that no campaigns are available, redirects the User to the homepage and ends the use case.
- 2.a: “The user does not apply a search filter”: the system skips to step 4, displaying the list of available campaigns already retrieved in step 1.
- 5.a: “The user decides not to request participation”: the system goes back to step 1.
- 6.a: “The user has already sent a participation request to this campaign”: the system displays a warning to the User.

Dictionary

*search filter: the criteria used by the Player to filter their search by campaign mode (online/offline) and name.

Discrepancies

- 1.a: the system does not display a warning message.

(Ivan Crescenzi)

“Create campaign”.

1. The User logs-in as DM.
2. The User enters campaign details* (name, time, frequency, start date, mode, platform if online* and city if offline*).
3. The User confirms the entered information.
4. The system displays players selection.
5. The User selects which players to invite to the campaign.
6. The User confirms the selection.
7. The system creates the campaign and displays a success message.
8. The system adds the new campaign to the list of available campaigns.

9. The system sends an invitation notification to players who have been selected by User.

Extensions

3.a “The entered data is not valid”. The system displays an error message and goes back to step 2.

5.a “The User does not select any player”. The system proceeds to step 7 and skips step 9.

7.a “A system or database error occurs during creation”. The system displays a failure notification, does not save the campaign and ends use case.

2.1 STORYBOARDS

Here are the following storyboards:

- Login (without info)
- Login (with info)
- Homepage (user not logged-in)
- Homepage (user logged-in)
- Character list
- Campaign list
- Single Campaign view

World Serpent Inn



Accedi

Email

mario.rossi@gmail.com



Password

La tua password



[Hai dimenticato la tua password?](#)

Accedi

OPPURE



Accedi con Google

Non hai un account? [Registrati qui](#)

World Serpent Inn



Accedi

Email

myemail@gmail.com



Password



[Hai dimenticato la tua password?](#)

Accedi

OPPURE



Accedi con Google

Non hai un account? [Registrati qui](#)

**World Serpent Inn**[Vedi campagne](#)[Vedi pg](#)[Crea campagna](#)[Crea pg](#)[Login](#)

Visualizza personaggi precostruiti

Visualizza una lista di personaggi precostruiti di questo sito e di personaggi creati dagli altri utenti

[Vai](#)

Inizia subito a giocare in una campagna



Visualizza campagne

Visualizza le campagne disponibili, e se vuoi unisciti ad una

[Vai](#)



Crea personaggio

Crea il tuo personaggio e mandalo al tuo master, oppure tienilo per te

[Vai](#)



Crea campagna

Inizia subito a creare la tua campagna, e scegli se invitare i tuoi giocatori preferiti o se aspettare che entri chi vuole (non preoccuparti, puoi sempre rifiutarlo)

[Vai](#)



World Serpent Inn

Vedi campagneVedi pgCrea campagnaCrea pg



▼ Player123

Logout

Visualizza personaggi precostruiti

Visualizza una lista di personaggi precostruiti di questo sito e di personaggi creati dagli altri utenti

Vai



Inizia subito a giocare in una campagna



Visualizza campagne

Visualizza le campagne disponibili, e se vuoi unisciti ad una

Vai



Crea personaggio

Crea il tuo personaggio e mandalo al tuo master, oppure tienilo per te

Vai



Crea campagna

Inizia subito a creare la tua campagna, e scegli se invitare i tuoi giocatori preferiti o se aspettare che entri chi vuole (non preoccuparti, puoi sempre rifiutarlo)

Vai

Pag. 14 a 29

CHARACTER LIST (Ivan Crescenzi)

**World Serpent Inn**

Vedi campagneVedi pgCrea campagnaCrea pgLogin

Gria



Una ragazza geniale e sempre allegra che però non ha avuto nessuno attorno che se ne accorgesse, ha studiato l'arcano a casa di sua zia e ora il destino l'ha portata a Barovia, voci di corridoio dicono che in realtà è un vampiro.

Livello 4
Umana, ranger uccisore di mostri

Caratteristiche principali: destrezza, intelligenza, saggezza

Specialità in combattimento:

- Tiro con l'arco
- Scoprire posizione e debolezze dei nemici
- Disseminare l'area di trappole inutili

 **SCARICA**

VerWohnt



Owlin nobile nato con poteri magici, dopo aver sconfitto un drago verde è stato posseduto da esso e ora non si sa che fine abbia fatto

Livello 7
Owlin, stregone di magia selvaggia

Caratteristiche principali: destrezza, carisma

Specialità in combattimento:

- Creare il caos
- Dare fuoco a tutto
- Utility varie

 **SCARICA**

Dill



Un Aasimar con la sindrome di Stendhal, figlio del dio dei sogni e della dea della natura, ha avuto tutto ciò che ha sempre desiderato, ora è diventato scrittore di romanzi fantasy

Livello 3
Aasimar, bardo/monaco

Caratteristiche principali: destrezza, carisma

Specialità in combattimento:

- Prendere a pugni più volte
- Creare illusioni letali con il suo "sussurri dissonanti"
- Cura e aiuto dei suoi alleati (se gli sono simpatici)

 **SCARICA**

Boone



Nato in un mondo corrotto, ha fatto un giuramento per creare un mondo migliore per i suoi figli, ora va in giro ad uccidere mostruosità nonostante la sua età avanzata

Livello 9
Umano, paladino giuramento degli antichi / guerriero m. di battaglia

Caratteristiche principali: forza, costituzione, carisma

Specialità in combattimento:

- Attaccare corpo a corpo con le sue due spade lunghe
- Cure e controllo di malattie e veleni per i suoi alleati
- Posizionamento della squadra grazie alle sue manovre

 **SCARICA**

Kyor



Venuto dall'anno 3318, è un agente segreto che opera in modo poco ortodosso. Cerca risposte sull'albero del mondo e sulla sua discendenza dopo che il suo villaggio è stato distrutto

Livello 13 (manuale 2024)
Aasimar, barbaro dell'albero del mondo

Caratteristiche principali: forza, costituzione

Specialità in combattimento:

- Fare male e non farsi male
- Lanciare alleati e nemici
- Vola

 **SCARICA**



Richiesta accettata ×
Complimenti, sei dentro
"Lost in the High Forest"!

Lista campagne disponibili



Tib Winterfield

The Winters Crest

Il re è qui, ha fatto una lista e non importa se sei buono o cattivo...

Seleziona campagna



Dungeon Monster

The tournament of tokens

C'è la possibilità di trionfare o di morire nel tentativo. Pronti?

Seleziona campagna



Outpost Owibear

Talisman of Dark Shadow

Volevi solamente un po' di riposo, ma i nani assetati di sangue non sono d'accordo....

Seleziona campagna



Finn Neuburger

The lambent delirium

Un'avventura "fatata" per i giocatori che amano il caos!

Seleziona campagna



Seth Harris

The Delian Tomb

Una tempesta di origini non naturali appropia la vostra carovana...

Seleziona campagna



Kat Kruger

X marks the spot

Una fuga dalla prigione si trasforma in una gara per vincere una reliquia.

Seleziona campagna

← Previous **1** 2 3 ... 67 68 Next →



"X marks the spot"

Dettagli

 Mensile/Venerdì - 09:00 PM

 Durata di una sessione: 3-4 h

 3/5 giocatori presenti 

 Roma/offline

Dungeon Master

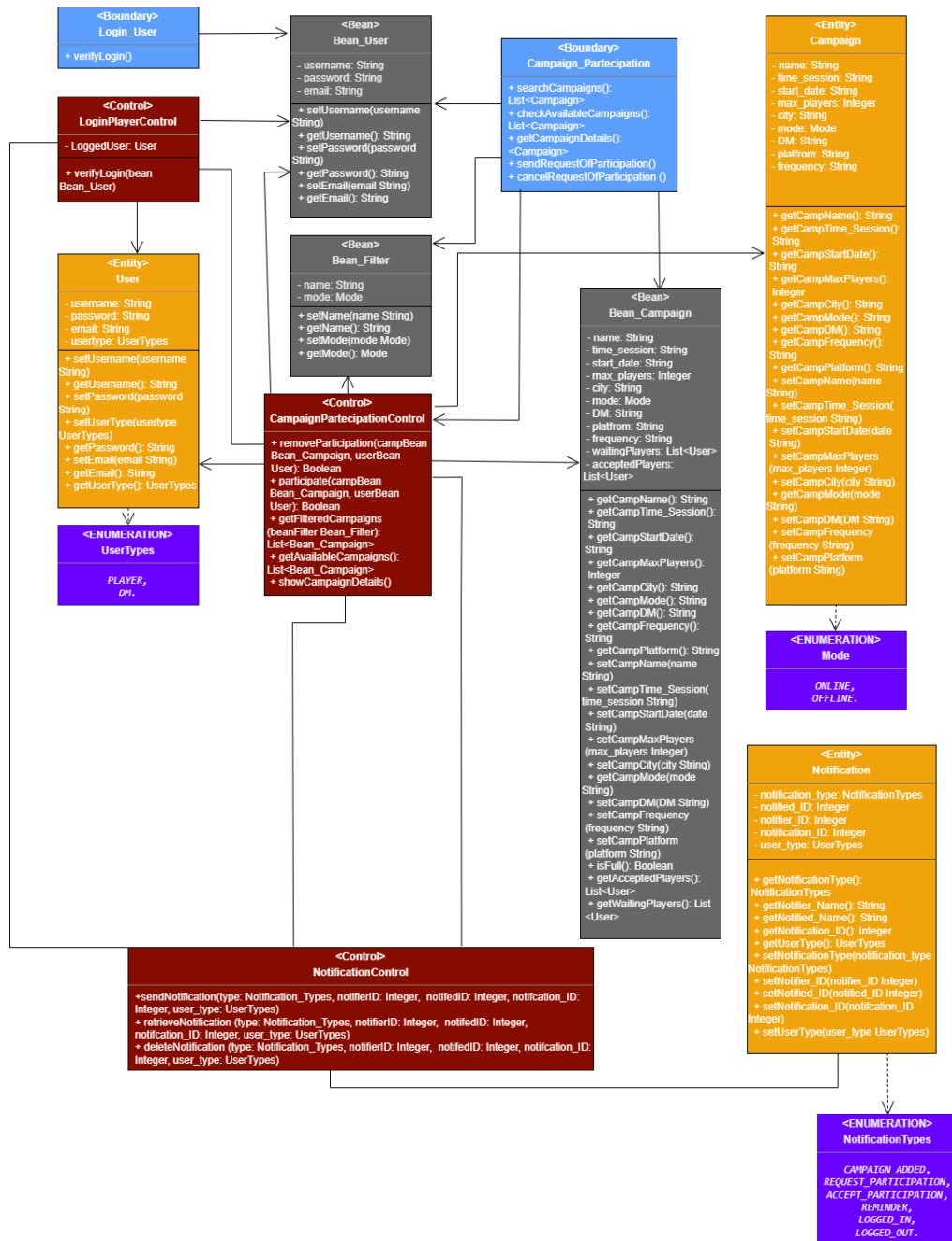
 Dungeon Monster 

Vuoi partecipare a questa campagna?

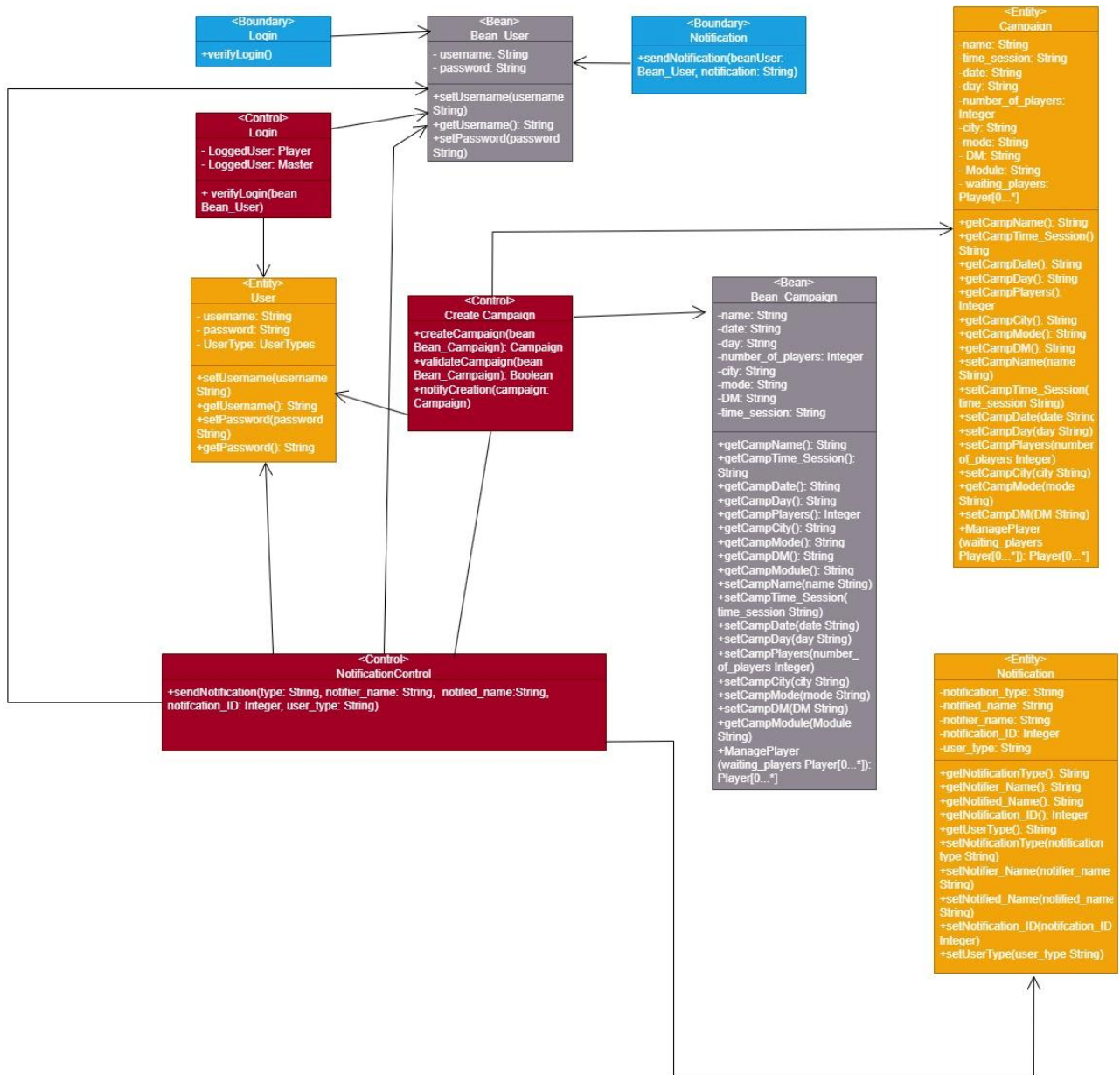
3.1 DESIGN – CLASS DIAGRAM

3.1.1 VOPC [Analysis]

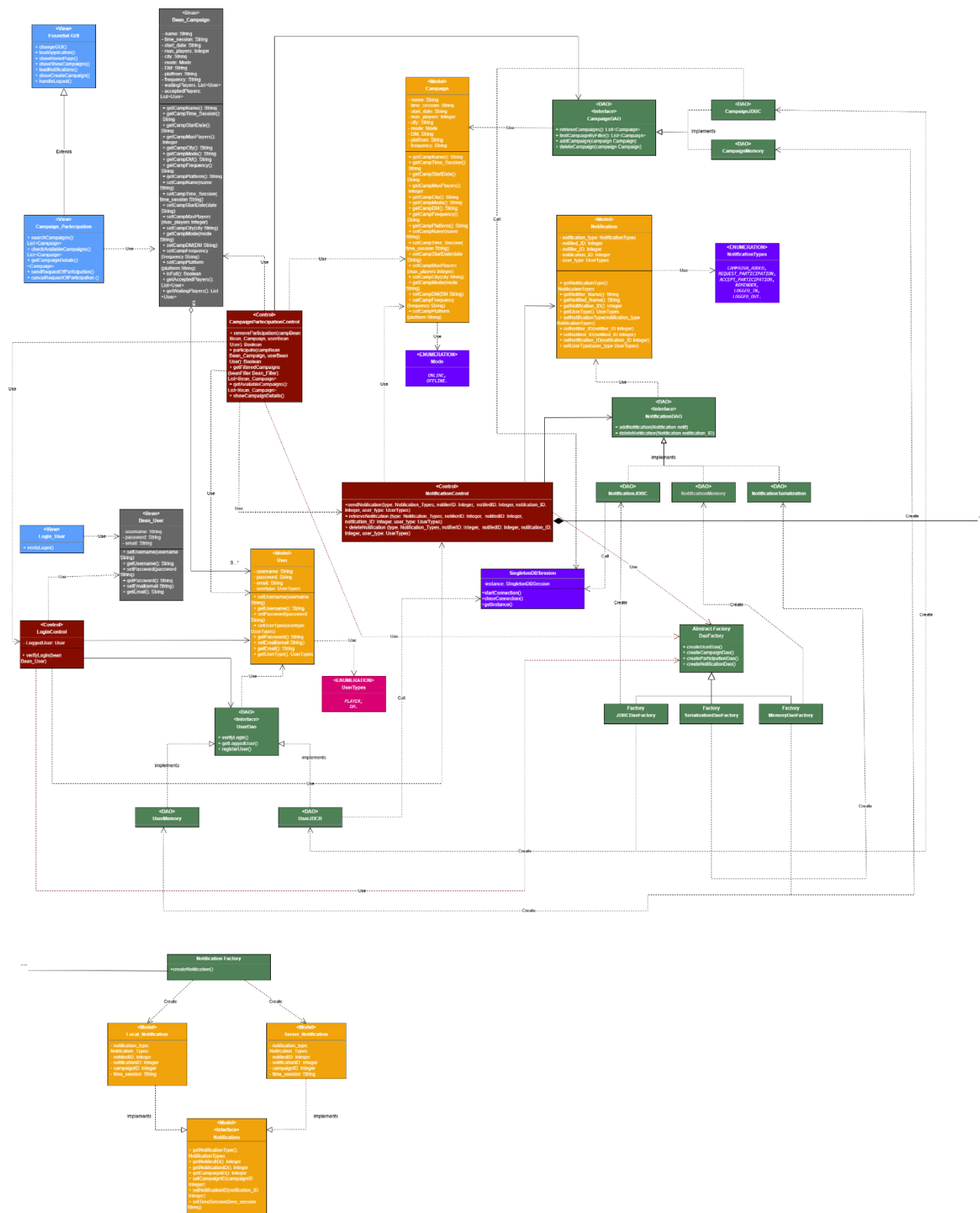
- VOPC (BCE principles) related to the use case “Plan campaign participation”.



- VOPC (BCE principles) related to the use case “Create campaign”.

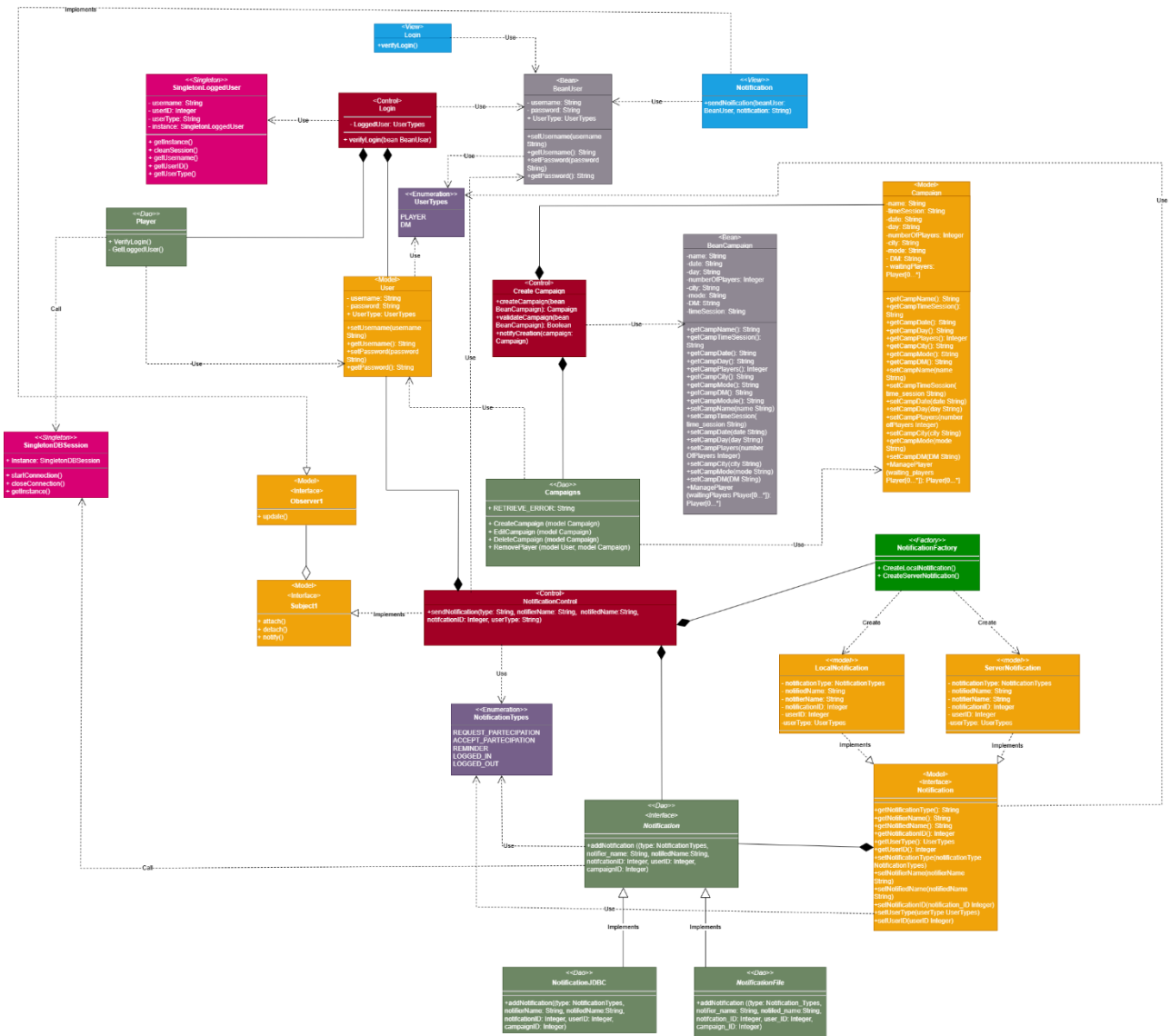


3.1.2.1 Abstract Factory pattern and Singleton Pattern in “Plan campaign participation” use case (Arianna Gabrieli)



Note: the red arrows used in the diagram are useful to highlight the relationship between the controllers and the abstract factory Daofactory.

3.1.2.1 Observer pattern and Singleton Pattern in “Create Campaign” use case (Ivan Crescenzi)



3.2 Design patterns

The project was developed by using several design patterns from the GoF classification to improve the separation of responsibility and reduce coupling between classes.

1. Abstract Factory pattern:

This pattern is implemented via the abstract class `DaoFactory` and its concrete subclasses, effectively encapsulating the creation logic of data access objects. In this way, it's possible to change the persistency layer

without modifying the source code of the services utilizing the DAO classes. Consequently, the architecture becomes more flexible and extensible.

2. Singleton pattern (Database Connection and User Session):

The Singleton pattern is implemented by SingletonDBSession and SingletonLoggedUser classes. This pattern ensures that a class has only one instance throughout the application's lifecycle, making sure that there is a global point of access to it.

For the database connection, this prevents creation of multiple concurrent sessions, which would waste system resources and degrade performance.

For the user session, this centralizes the state of the logged-in user session across the application.

3. Observer pattern:

The Observer pattern is implemented by NotificationControl class. This pattern ensures that whenever the underlying data state changes (such as adding or removing a notification), all dependent views are automatically notified and updated. In this way, it establishes a loosely coupled architecture where the business logic remains completely independent from the user interface, making the application easier to maintain and scale.

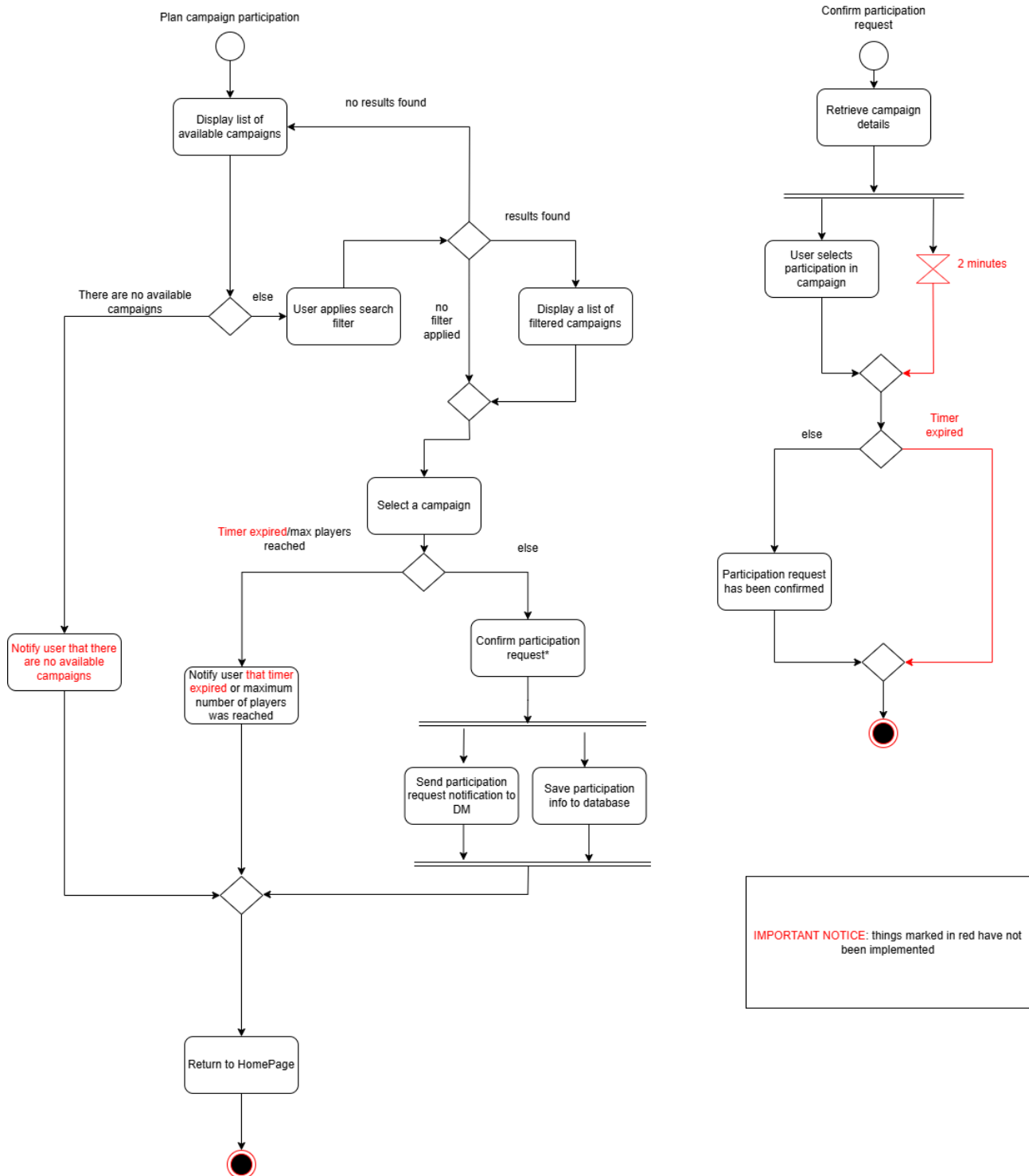
4. Factory method pattern:

The Factory method pattern is used to define a common interface for creating objects, to instantiate either the LocalNotification class or the ServerNotification class without binding the client code to one of these specific implementations.

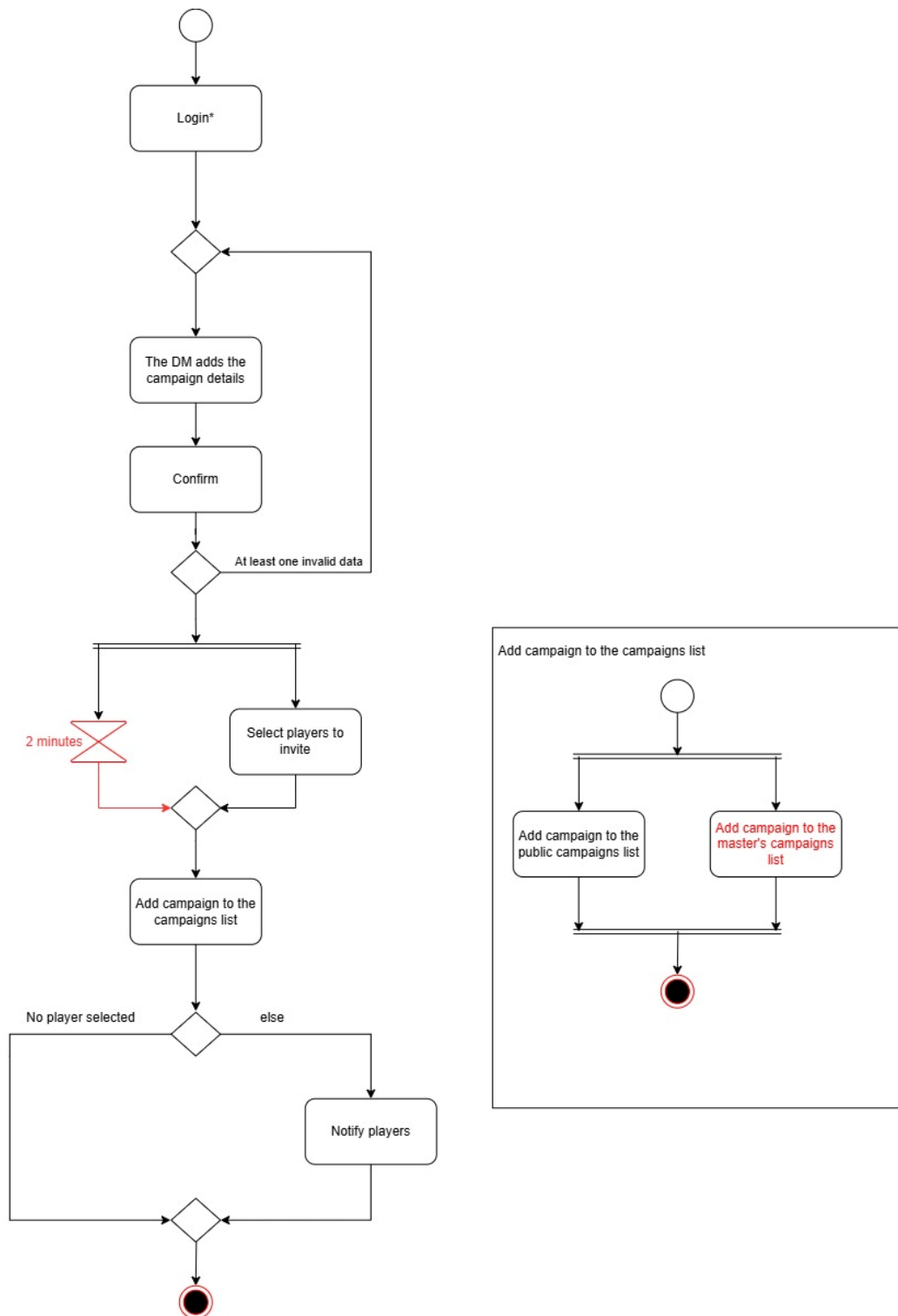
This way, the addition of a new notification is very straightforward and code extensibility is improved.

3.3 Activity Diagram

3.3.1 Activity diagram (Use case “Plan campaign participation”, Arianna Gabrieli)

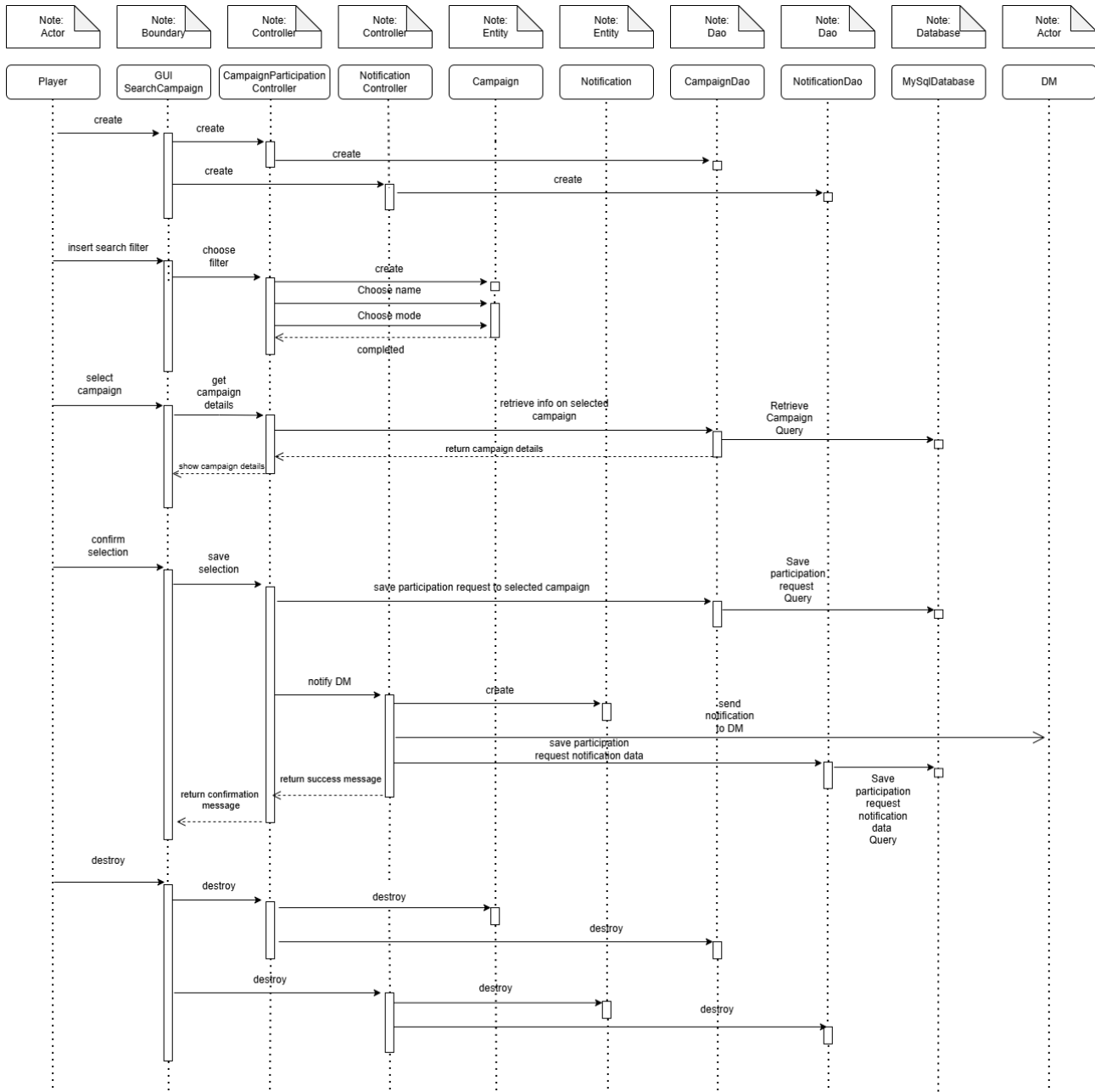


3.3.2 Activity diagram (Use case “Create Campaign”, Ivan Crescenzi)

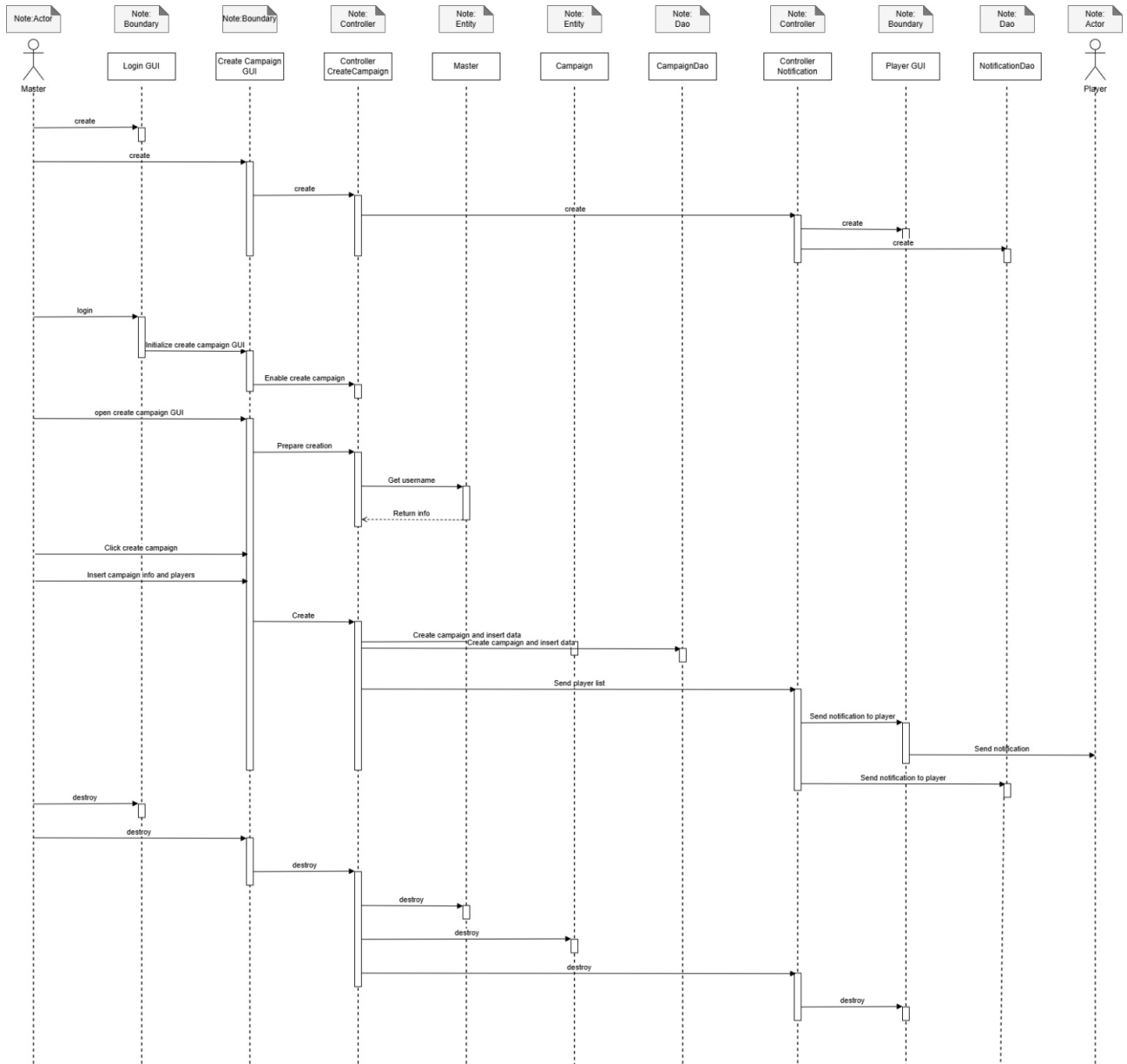


3.4 SEQUENCE DIAGRAM

3.4.1 Sequence diagram (Use case “Plan campaign participation”, Arianna Gabrieli)

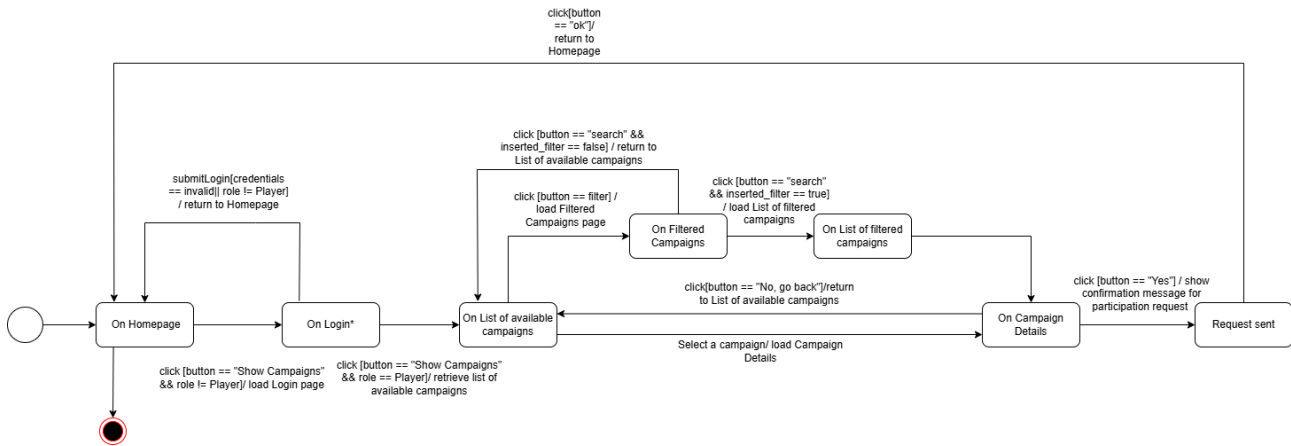


3.4.2 Sequence diagram (Use case “Create campaign”, Ivan Crescenzi)

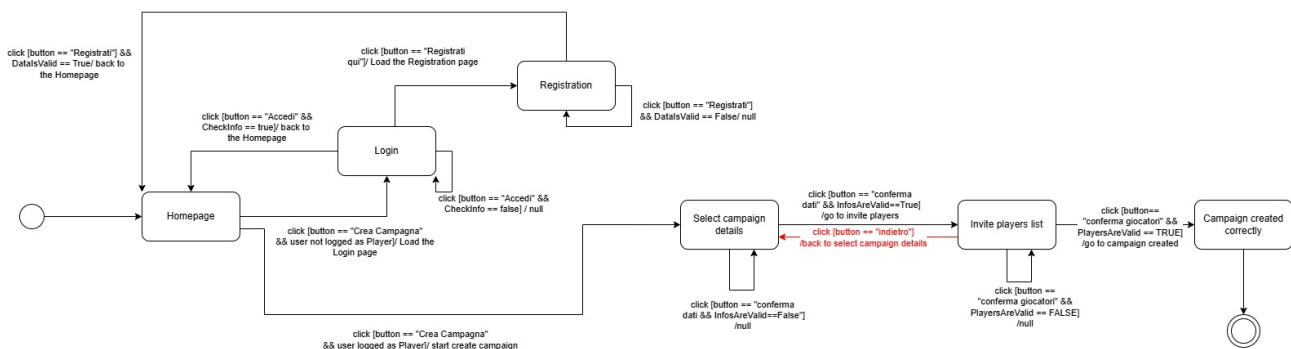


3.5 STATE DIAGRAM

3.5.1 State diagram (Use case “Plan campaign participation”, Arianna Gabrieli)



3.5.2 State diagram (Use case “Create Campaign”, Ivan Crescenzi)



4 TESTING

4.1 LoginControl Test (Arianna Gabrieli)

- **Authentication:** verifies successful user authentication when user credentials are provided and it makes sure that login fails when credentials are invalid.
- **Registration and input validation:** checks if the system detects when a username is already taken and if `TextTooLongException`, `TextTooShortException`, `InvalidValueException` are thrown when entering invalid input.

4.2 UsernameTaken Test (Arianna Gabrieli)

- It verifies if UsernameTaken is thrown when attempting to register a user with a username that already exists in the system.

4.3 CampaignParticipationContol Test (Arianna Gabrieli)

- **Participation request:** verifies if a participation request to a specific campaign is correctly stored in the persistence layer and sent. In addition, confirms that an existing campaign participation request can be successfully deleted from the system.

4.4 NotificationControl Test (Ivan Crescenzi)

- It verifies that the campaign is created after the campaign info are provided (necessary for the next step).
- It verifies that after successful creation of a campaign, a notification is correctly generated and sent to the selected player.

4.5 Request Already Sent (Ivan Crescenzi)

- It ensures that a RequestAlreadySent exception is thrown if a user attempts to submit another participation request to the same campaign.

5 DATABASE

The system can save notifications in a binary file DbNotification.dat with serialization.

The other type of DBMS in the system is the connection to a MySQL server to access the information.

6 CODE & SONARCLOUD

Link to GitHub page with project repository:

https://github.com/GabCresc/TheWorldSerpentInn_project

Link to SonarCloud page:

https://sonarcloud.io/project/overview?id=GabCresc_TheWorldSerpentInn_project

7 VIDEO

Link to project video:

<https://drive.google.com/file/d/197Yf2w80zWJed3n5OY-wycmMJebrO7hG/view>